



Solving and Graphing Linear Inequalities

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Learning Objectives

- Solve one- and multi-step linear inequalities
- Reverse the inequality sign when multiplying or dividing by a negative
- Graph solution sets on a number line
- Express solutions in interval notation

Solve each inequality completely, showing all steps, then write the solution in interval notation.

1. Solve and graph on a number line:

$$3x - 5 < 7$$

Answer: _____

2. Solve. Watch the sign when dividing by a negative:

$$-2x + 1 \geq 9$$

Answer: _____

3. Solve:

$$5x + 3 \leq 2x - 6$$

Answer: _____

4. Solve:

$$\frac{x}{3} - 2 > 1$$

Answer: _____

5. Solve. Remember to flip the sign:

$$4 - 3x > -8$$

Answer: _____

6. Solve:

$$2(x - 1) \leq 3x + 4$$

Answer: _____



7. Solve:

$$\frac{2x - 1}{4} \geq 3$$

Answer: _____

8. Solve:

$$-5 \leq \frac{x}{2} + 1$$

Answer: _____

9. Solve:

$$7 - 2x \leq 4x - 11$$

Answer: _____

10. Solve:

$$-3(x + 2) > 9$$

Answer: _____





Key reminder: flip the inequality symbol whenever you multiply or divide both sides by a negative number.

Solutions

1. Solve and graph on a number line:

$$3x - 5 < 7$$

→ Add 5 to both sides: $3x < 12$.

→ Divide by 3: $x < 4$.

→ Open circle at 4, shade left.

Answer: $x < 4 \Rightarrow (-\infty, 4)$

2. Solve. Watch the sign when dividing by a negative:

$$-2x + 1 \geq 9$$

→ Subtract 1: $-2x \geq 8$.

→ Divide by -2 and FLIP: $x \leq -4$.

→ Closed circle at -4, shade left.

Answer: $x \leq -4 \Rightarrow (-\infty, -4]$

3. Solve:

$$5x + 3 \leq 2x - 6$$

→ Subtract 2x: $3x + 3 \leq -6$.

→ Subtract 3: $3x \leq -9$.

→ Divide by 3: $x \leq -3$.

Answer: $x \leq -3 \Rightarrow (-\infty, -3]$

4. Solve:

$$\frac{x}{3} - 2 > 1$$

→ Add 2: $x/3 > 3$.

→ Multiply by 3: $x > 9$.

Answer: $x > 9 \Rightarrow (9, \infty)$

5. Solve. Remember to flip the sign:

$$4 - 3x > -8$$

→ Subtract 4: $-3x > -12$.

→ Divide by -3 and FLIP: $x < 4$.

Answer: $x < 4 \Rightarrow (-\infty, 4)$



6. Solve:

$$2(x - 1) \leq 3x + 4$$

→ Distribute: $2x - 2 \leq 3x + 4$.

→ Subtract $2x$: $-2 \leq x + 4$.

→ Subtract 4: $-6 \leq x$.

Answer: $x \geq -6 \Rightarrow [-6, \infty)$

7. Solve:

$$\frac{2x - 1}{4} \geq 3$$

→ Multiply by 4: $2x - 1 \geq 12$.

→ Add 1: $2x \geq 13$.

→ Divide by 2: $x \geq 13/2$.

Answer: $x \geq \frac{13}{2} \Rightarrow [\frac{13}{2}, \infty)$

8. Solve:

$$-5 \leq \frac{x}{2} + 1$$

→ Subtract 1: $-6 \leq x/2$.

→ Multiply by 2: $-12 \leq x$.

Answer: $x \geq -12 \Rightarrow [-12, \infty)$

9. Solve:

$$7 - 2x \leq 4x - 11$$

→ Add $2x$: $7 \leq 6x - 11$.

→ Add 11: $18 \leq 6x$.

→ Divide by 6: $3 \leq x$.

Answer: $x \geq 3 \Rightarrow [3, \infty)$

10. Solve:

$$-3(x + 2) > 9$$

→ Distribute: $-3x - 6 > 9$.

→ Add 6: $-3x > 15$.

→ Divide by -3 and FLIP: $x < -5$.

Answer: $x < -5 \Rightarrow (-\infty, -5)$

